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Earth and Planetary Science Letters 179 (2000) 325–334

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Late Silurian paleomagnetic pole from the Holy Cross Mountains: constraints for the post-Caledonian tectonic activity of the Trans-European Suture Zone

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Received 14 March 2000; received in revised form 17 April 2000; accepted 17 April 2000

Abstract

In spite of extensive studies, the tectonic evolution of the most prominent geological boundary in Europe, the Trans-European Suture Zone (TESZ), remains enigmatic. End-member models are either autochthonous, requiring in situ Phanerozoic reworking of the pre-Cambrian crust, or allochthonous, including accretion of mobile Caledonian and Variscan terranes. Continent-scale Variscan right-lateral displacements along the TESZ have been also postulated. A paleomagnetic study of Upper Silurian diabases from the southern part of Holy Cross Mts. (central Poland) was undertaken to test between these models. A primary characteristic magnetization was isolated in these rocks. It passes the fold test and therefore is considered to be Late Silurian in age. The corresponding pole (12°S, 340°S) is concordant with the Ludlovian segment of the apparent polar wander path for Baltica. This result does not support hypotheses about significant post-Caledonian dextral tectonic movements along the TESZ. Tectonic movements and accretion of blocks along the SW edge of Baltica must have occurred before the latest Silurian. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: suture zone; Swiety Krzyz Mountains; Upper Silurian; diabase; paleomagnetism

1. Introduction

The most prominent geological boundary in Europe is the Trans-European Suture Zone (TESZ), which separates a set of Phanerozoic terranes in the south and west from the pre-Cambrian East European Craton [1,2]. The TESZ forms the boundary between the pre-Cambrian lithosphere of the Baltic shield and East European Craton, characterized by a thick crust (c. 45 km), low heat flow and a tectonothermal age of ca. 3000 to 800 Ma, and the Paleozoic mobile belts of western Europe, with a thinner crust (ca. 30 km), higher heat flow and an age of 450 to 290 Ma [3–5]. The TESZ passes from the North Sea in the north-west to the Black Sea in the south-east through central Europe (Fig. 1a). Its northern segment is regarded as the Caledonian collision zone marked by overthrusting of Lower Paleozoic strata onto Baltic crystalline basement [6–8]. Some authors have proposed

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PII: S0012-821X(00)00125-4

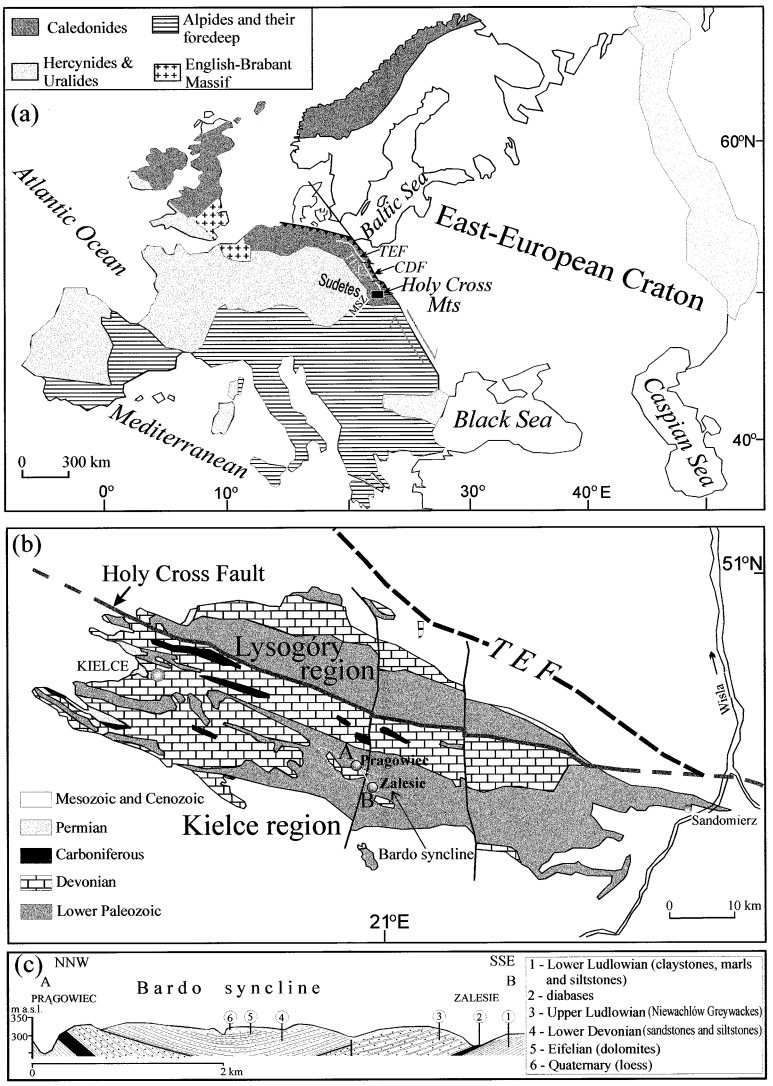


Fig. 1. (a) The location of the Holy Cross Mts. within a tectonic sketch map of Europe. Following some authors (see text) the Trans-European Suture Zone (TESZ) is marked here as a Variscan dextral transform fault zone. The Trans-European Fault (TEF) follows the East European Platform border covered by the Lower Paleozoic strata deformed and overthrusted during the Silurian. Their maximum extent is marked as the Caledonian Deformation Front (CDF). MSZ, Moravo–Silesian zone. (b) Geological sketch map of the Holy Cross Mts. with sampling localities. (c) Simplified geological section through the Bardo syncline.

that large-scale strike-slip movements took place on the SW edge of the East European Craton, along the TESZ in Early Paleozoic time or during the Variscan orogenic cycle. Brochwicz-Lewinski et al. [9] interpret the SW boundary of the East European Craton as a sinistral strike-slip fault that was active from the Cambrian to the Early Devonian and created an eastward displacement of some Caledonian terranes right to the southern foreland of Baltica. In contrast a significant (~300–350 km) Variscan right-lateral displacement along the TESZ has been postulated by Pegrum [10]. Right-lateral motions of the Hercynide zone along the TESZ and other parallel shear zones have also been hypothesized by Arthaud and Matte [11], Dewey [12], Neugebauer [13]

and Matte et al. [14]. Badham [15] presented a strike-slip model for the western European Hercynides. In this interpretation the so-called 'Tornquist line' was of fundamental importance for the Variscan strike-slip processes.

The Holy Cross Mountains (HCM) located in the vicinity of the East European Craton, provide a good area to test the possibility of significant strike-slip displacement along the TESZ. Available Devonian and Ordovician paleomagnetic poles from the southern part of the HCM [16] deviate significantly from the Baltic apparent polar wander path and seem to support the model of Badham [15], with Lewandowski [16] proposing a significant ($\sim$1000 km) Variscan dextral strike-slip displacement of the southern region of HCM along the SW border of the East European Craton. The Variscan time of that movement and, particularly, the quality of Devonian paleomagnetic data have been contested by Nawrocki [17]. Basic for Lewandowski's model [16] are the paleomagnetic directions DS1 and DS2, which are parallel to the axes of studied folds; their precision parameter k does not exceed 8.

This paper presents a new paleomagnetic pole that clearly defines the pre-Variscan (Late Silurian/Early Devonian) paleoposition of the southern HCM. The results are important because the area studied is the best recognized and exposed Paleozoic massif located in a critical area of the Paleozoic Platform of central Europe i.e. in the immediate vicinity of the East European Platform border. A transcontinental displacement of this relatively small tectonic block should be paleomagnetically detectable because the Late Silurian–Early Devonian segment of the reference Baltic apparent polar wander path is well defined.

2. Geology

The Paleozoic core of the HCM, including sedimentary rocks ranging in age from Upper Proterozoic to Lower Carboniferous, emerged from below the Permian and Mesozoic cover during the Late Cretaceous movements [18]. It consists of two distinct tectono-stratigraphic units separated by the Holy Cross Dislocation: the southern Kielce region and the northern Lysogóry region (Fig. 1b). The Kielce region structurally belongs to the Malopolska Massif [19], regarded by some authors [20] as the first Gondwana-derived lithospheric plate rifted in the Cambrian and accreted to the margin of Baltica. However, this hypothesis is not supported by the record of Cambrian fossils from both regions of HCM. They are taxonomically homogeneous and consistent with the fossil record of Scandinavia, the East European Platform and Upper Silesia [21]. Early Paleozoic proximity of the Małłopolska Massif and East European Platform can also be inferred from the Baltic (1721 Ma) age of muscovites recovered from the Middle Cambrian of the southern region of HCM [22]. From the Cambrian to the Devonian both units of the HCM underwent different tectonic and facies development [23,24]. Two pre-Variscan tectonic events were recognized in the Kielce region: one between the Middle Cambrian and the Early Ordovician, and one between the Late Silurian and the Early Devonian. In the Lysogóry region, the folded rocks (Middle Cambrian to Earliest Carboniferous) comprise a complete sedimentary sequence without any distinct breaks [23]. The post-Variscan paleomagnetic data (see e.g. [16]) indicate that after the Variscan folding, the HCM became a relatively stable block with respect to Baltica.

The Bardo syncline is located in the central part of the Kielce region. A diabase intrusion occurs in Silurian rocks of this syncline, close to the stratigraphic boundary between Lower Ludlovian graphtolite shales and Upper Ludlowian greywackes (Fig. 1c). The overall thickness of the intrusion varies between 20 and 30 m. Preliminary tectonic and stratigraphic observations point to a Late Ludlowian–Siegenian age interval for the Bardo intrusion. The Ludlowian strata along with piercing diabases were folded and discordantly covered by Lower Devonian (Emsian) sediments, equivalent to the Old Red facies [25]. In the vicinity of Pragowiec the value of the unconformity is 30–50° [26]. Recent results of studies of the Late Silurian–Early Devonian sequences from the southern part of HCM show that the main phase of late Caledonian movements occurred here in the latest Ludlowian [27,28] or Late Ge-

Table 1
Mean directions and paleopole for components '*S*' and '*Z*' from Bardo diabases

Locality (coordinates)	*a*/*d*	Comp.	Site	*N*/*N**	In situ			Bedding corrected			Lat.	Long.	dp	dm
					Dec/Inc	*k*	α_{95} (°)	Dec/Inc	*k*	α_{95} (°)				
Zalesie (50.7°N, 20.9°E)	20°/38°	*S*	1	9/8	36/9	53.1	7.7	38/−30	53.2	7.7				
		S	2	9/6	44/7	242.3	4.3	47/−30	225.5	4.5				
		S	3	10/10	38/3	120.9	4.4	42/−35	138.5	4.1				
		Z	1+2+3	28/13	46/33	80.6	4.6	41/−4	83.8	4.6	27°S	333°E	2	5
Pragowiec (50.8°N, 20.8°E)	205°/52°	*S*	1	11/11	87/−74	142.7	3.8	41/−31	145.4	3.8				
		S	2	9/4	96/−73	82.0	10.2	44/−33	95.1	9.5				
Zalesie and Pragowiec		*S*	samples	48/39	46/−24	3.8	13.6	41/−32	100.3	2.3	12°S	340°E	1	3
		S	sites	5/5	47/−25	3.1	50.5	42/−32	505.4	3.4	12°S	340°E	2	4

a/*d*, azimuth and magnitude of dip of surrounding Silurian sedimentary rocks; *N*, number of samples measured; *N**, number of samples used in calculation of mean; Dec/Inc, declination/inclination (in degrees); *k*, precision parameter (after [44]); α_{95}, semi-angle of the cone of 95% confidence; Lat., latitude of south paleomagnetic pole; Long., longitude of south paleomagnetic pole; dp, error of the distance between site and paleopole; dm, paleodeclination error.

dinian [29]. Hence, the age of the Bardo diabases can be constrained between Late Ludlowian and Early Gedinian. Przybylowicz and Stupnicka [30] correlate the Upper Ludlowian Niewachlów greywackes with the diabase sills. According to their opinion the volcanic material from greywackes and diabases originated from the same magmatic center. Additionally, the close spatial relationship between the diabase sills and locations of maximum thicknesses of greywackes containing subvolcanic material points to the Late Ludlovian age of the diabases [30].

3. Sampling

Only two exposures of the diabase intrusion are available in the HCM. They are situated on the opposite limbs of the Bardo syncline (Fig. 1c), which allowed me to perform a paleomagnetic fold test. Three sites consisting of 28 samples were collected near the village of Zalesie, located on the southern limb of the Bardo syncline. Two sites consisting of 20 samples were collected in the Pragowiec Ravine near the village of Bardo, on the northern limb of the Bardo syncline. In both localities the earlier described [25] contact of diabase bodies with Silurian sedimentary rocks is not exposed at present. The bedding (strike and dip) of Silurian shales and greywackes was measured in several sites located in the vicinity of studied outcrops. A small (4°) scatter of structural data was observed and therefore the mean values of the bedding were used for tectonic correction of paleomagnetic data. These rocks dip 38° towards NNE in the nearest surroundings of the intrusion near Zalesie and 52° towards SSW in the vicinity of intrusion from the Pragowiec Ravine (Table 1).

4. Paleomagnetic results

Pilot samples were subjected to both alternating field (AF) and thermal demagnetization experiments. AF was more effective and thus, the majority of samples was demagnetized using this technique. Demagnetization results were analyzed using orthogonal vector plots [31], and the directions of the linear segments were calculated using principal component analysis [32]. The paleomagnetic fold test of McFadden [33] was used to assess origin of the characteristic high-coercivity component. Magnetic mineralogy was determined from isothermal remnant magnetization (IRM) techniques, thermomagnetic analyses, and scanning electron microscope (SEM) images. Anisotropy of magnetic susceptibility (AMS) was measured on a KLY-2 susceptibility bridge and

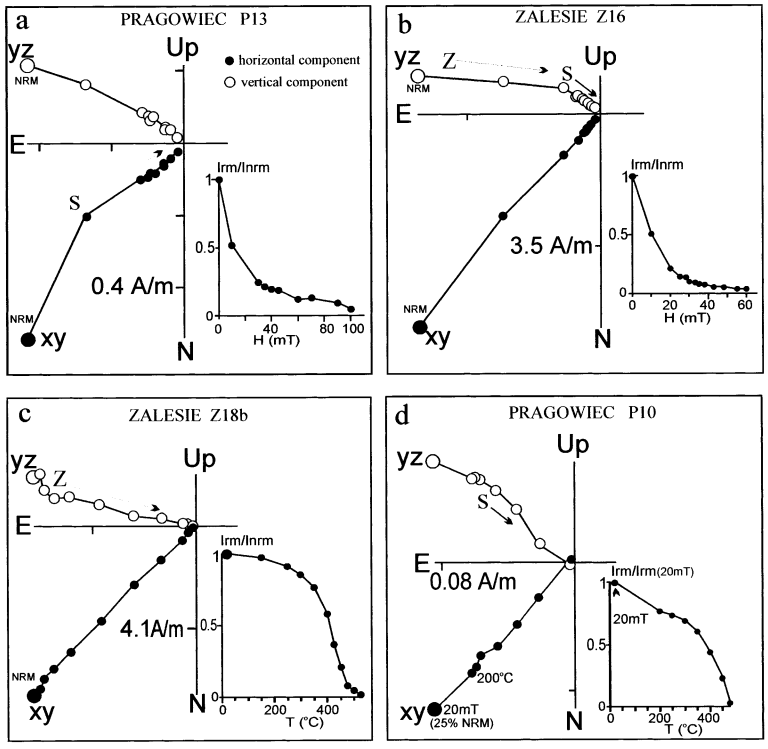


Fig. 2. Typical demagnetization characteristics (orthogonal plots and intensity decay curves) of Bardo diabases from Pragowiec and Zalesie; results of AF (a, b), thermal (c) and mixed (d) cleaning. Open circles in the orthogonal plots represent vertical projections, filled circles represent horizontal projections. The orthogonal plots are presented in tilt-corrected coordinates.

resulting AMS fabrics were computed using the ANISO program [34]. Measurements of the AMS revealed a low degree of magnetic grain alignment (1.2–3.8%).

The natural remnant magnetization (NRM) intensities of diabases from Zalesie range between 4.2 and 10.6 A/m. The intensities of the diabases from Pragowiec are between 0.3 and 1.1 A/m.

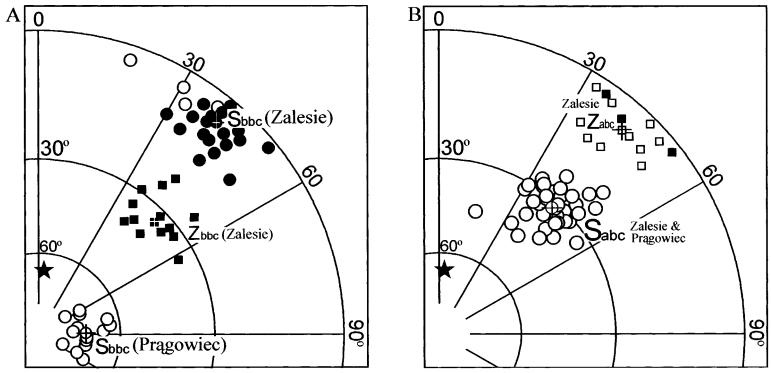


Fig. 3. Stereographic projections of line-fit directions *S* (circles) and *Z* (squares) from Bardo diabases presented before (A) and after tectonic correction (B). Open (closed) symbols denote upward (downward) pointing inclinations. Crossed symbols show the mean directions based on samples. Stars on the stereonets point the local present-day field direction.

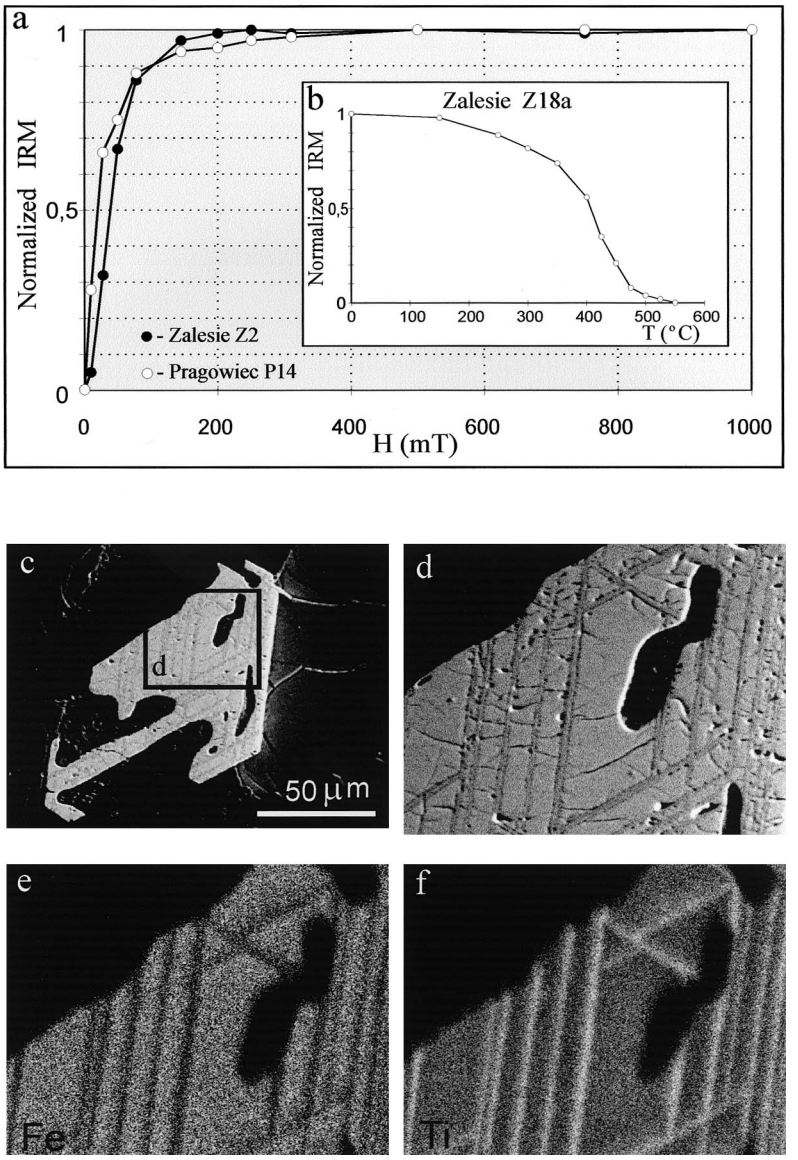


Fig. 4. (a) IRM acquisition curves for diabase samples from Pragowiec and Zalesie. (b) Thermal demagnetization of IRM obtained for diabase sample from Zalesie. (c) SEM image of polished surface of diabase sample Z10 from Zalesie, showing titanomagnetite grain with ilmenite intergrowths. (d) SEM image of selected area of the same grain. (e) Map of distribution of Fe content. (f) Map of distribution of Ti content.

Most of the samples revealed a low-coercivity NRM component removed by alternating fields up to 20 mT and temperatures of about 520°C (Fig. 2a–c). In the samples from Pragowiec this component was strongly dispersed and therefore was not evaluated statistically. However 13 samples from Zalesie contained well-clustered low-coercivity component (Z) defined by best-fitted lines (Figs. 2b,c and 3; Table 1). After removal of the low-coercivity overprint above 20 mT a high-coercivity component S can be identified in the samples from both localities (Fig. 2). The S magnetization is isolated as a straight-line segment directed towards the origin in orthogonal projec-

tion. Identified in all five sites from both localities, it yields an overall mean direction of 047°/−25°, $k = 3.1$, $\alpha_{95} = 50.5°$ in situ, and 042°/−32°, $k = 505.4$, $\alpha_{95} = 3.4°$ after bedding correction (Table 1, Fig. 3). The improvement in clustering of the site mean direction after bedding correction is significant at the 95% level of confidence using the fold test of McFadden [33]. The critical value of distribution at this level of confidence is 3.57 whereas the value of diagnostic parameter SCOS1 is 0.356. Optimal clustering of the direction *S* is observed after full bedding correction and there is no suspicion of synfolding remagnetization.

IRM experiments and subsequent thermal demagnetization of the saturation magnetization confirm the predominance of magnetic carriers from the ulvospinel-magnetite solid solution series (Fig. 4a,b). The structure of NRM points to the presence of two magnetic phases differing in coercivity. Thermal and AF demagnetization of NRM indicate that the *S* component is carried by moderately high-coercivity minerals with unblocking temperature of about 450°C (Figs. 2a,b,d and 4a). This demagnetization behavior is typical for basaltic lava containing deuterically oxidized titanomagnetite [35–38]. The exsolution process involves a change in grain size, grain shape, and composition of the titanomagnetite. The changes will generally involve a decrease in grain size, and will therefore tend to increase the coercivity (see [39], p. 410). Observation of SEM images supports this interpretation. Magnetite–ilmenite intergrowths are common in the studied rocks (Fig. 4c–f).

The pre-folding *S* direction was converted to the paleomagnetic pole. The paleopole is compared to the apparent polar wander path (APWP) for Baltica, constructed using spherical spline methods incorporated in the GMAP program of Torsvik and Smethurst [40] and paleomagnetic data set published by Torsvik et al. [41]. The paleopole *S* from the Holy Cross Mts. corresponds to the Ludlovian (421–414 Ma) segment of the Baltic APWP (Fig. 5).

The low-coercivity component *Z* from Zalesie was also converted to the paleomagnetic pole. It fits well to the Early Carboniferous segment of

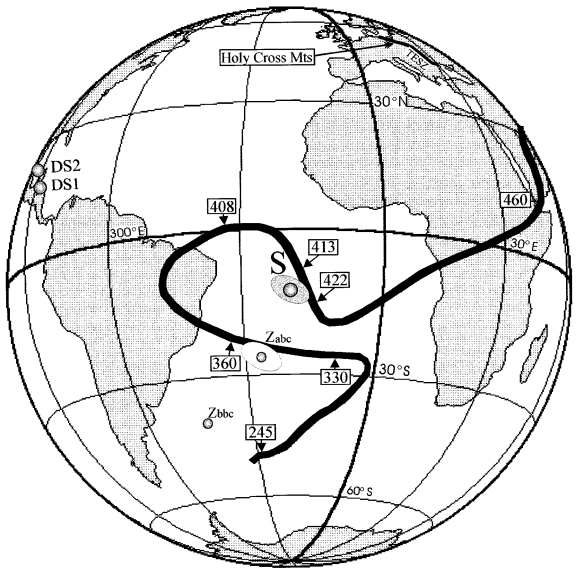


Fig. 5. Apparent south polar wander path of Baltica during the Paleozoic (for construction details see text) and the Late Silurian paleopole (labeled *S*) from this paper. Paleopole calculated for the low-coercivity component *Z* in present (Zbbc) and bedding-corrected (Zabc) coordinates, and paleopoles DS1 and DS2 obtained [16] from the Lower Devonian clastic rocks of Holy Cross Mts. are also presented. The ages adopted from the time calibration of Palmer [43] are shown in Ma.

the mentioned APWP after correction for the Variscan deformation. It deviates significantly from the APWP in in situ coordinates (Fig. 5).

5. Discussion and conclusions

The low-coercivity component *Z* from Zalesie reflects most probably an Early Carboniferous overprint. It may be related to the Variscan tectonic activity that caused the final folding of the Paleozoic strata in the Early Namurian [27].

The positive result of the fold test, the lack of symptoms of regional heating of surrounding sedimentary rocks [42], and the nature of magnetic minerals document a primary (thermal) or near primary (thermochemical) origin of the component *S*. Its pre-folding origin implies an age older than the Late Caledonian deformations, i.e. the latest Ludlovian [27,28] or Late Gedinian [29]. Taking into consideration the spatial rela-

tionship between the diabases and sedimentation of synorogenic Niewachlów greywackes [30] a Late Ludlovian age for the diabases and primary characteristic remanence is most probable.

Slight uncertainties in age do not influence the geotectonic conclusions. The Late Silurian paleomagnetic pole *S* from the Holy Cross Mts. does not support the hypothesis of large-scale post-Caledonian dextral tectonic movements along the Trans-European Suture Zone (e.g. [10,15,16]) and favors an in situ model for the Variscan evolution of the Holy Cross Mts. It also does not indicate any local clockwise tectonic rotations of the investigated rocks. Moreover, the pre-folding origin of the paleopole *S* indicates that the Late Caledonian deformation in the southern part of the Holy Cross Mts. was not accompanied by such tectonic movements. Because of the shape of the border of Baltica, a dextral strike-slip movement along the TESZ would have created clockwise rotation of the Malopolska terrane (see [16]). A small ($\sim$15°) post-Silurian tectonic rotation of this unit, anticlockwise, cannot be excluded if the characteristic component *S* would be of Late Pridolian age. This rotation would cancel the deviation of paleomagnetic pole from the Late Pridolian segment of the Baltic APWP and could for example be created by a slight strike-slip displacement of the southern part of the Holy Cross Mts. along the Holy Cross Fault (see e.g. [45]). This solution however is only hypothetical since the Late Ludlovian age of the characteristic component *S* is more likely.

The paleopole *S* from the Bardo diabases supports the thesis that the main tectonic movements and accretion of blocks plastered along the SW edge of Baltica took place before the latest Silurian. According to Unrug et al. [45] the Malopolska terrane docked onto Baltica during the Late Silurian diachronous closing of the Tornquist sea. This interpretation refers to the model of Soper et al. [46] but is not in agreement with the reconstruction of Torsvik et al. [47] that favors mid-Ordovician closure of the Tornquist sea. The paleopole *S* from the Holy Cross Mts. is younger than the above-mentioned processes and therefore can not help to solve this problem.

This study demonstrates that poor-quality paleomagnetic data should not be used as critical evidence for construction of any paleotectonic models. Especially the reliability criterion defining sufficient values of statistical parameters (see [48]) should not be ignored. In spite of poor statistic parameters the paleopoles DS1 and DS2 from the Holy Cross Mts. (Fig. 5) were used for construction of the model of transcontinental displacement [16] contested in this paper. The Eastern Variscan fold belt and its foreland provided another example of a geodynamic interpretation based on poor quality paleomagnetic data. Paleomagnetic studies of the Silurian ophiolitic belt from the eastern part of Sudetes yielded a well-defined paleopole consistent with the Silurian segment of the Baltic APWP [49] whereas investigations of the Devonian rocks from the Moravo–Silesian zone gave a dispersed ($\alpha_{95} = 16.3$; $k = 17.9$) Late Devonian paleodirection with declination different by about 90° from the Baltic coeval declination [50]. According to the authors (op. cit.) this paleodirection implies oroclinal bending of the eastern Variscan fold belt around the northern and eastern flanks of the Bohemian Massif. Recent geological interpretation (e.g. [51]) does not explain the Moravo–Silesian zone as a part of Rhenohercynian orocline but rather as a typical shear zone.

Acknowledgements

I thank S. Zbroja, Z. Migaszewski, Z. Kowalczewski and S. Salwa for their help in the field and for discussions on the local geology, and Leszek Krzemiński for microscopic analyses. Special thanks are expressed to Conall Mac Niocaill and John Stamatakos for critical review of the manuscript. Comments by Rob Van der Voo helped considerably with presentation of the final manuscript. This is a contribution to the research project 'Paleozoic Accretion of Poland' (PAP).*[RV]*

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